PROJECT PROPOSAL • SIMULATION STUDY • 3-PERSON TEAM

IoT Device Behaviour Fingerprinting and Simulation-Based Anomaly Detection

A focused two-day simulation project combining IoT, cybersecurity, statistical analysis, and data visualization.

Project Type	Simulation-based IoT cybersecurity project
Team Size	3 members
Duration	2 intensive working days
Implementation	Python + NumPy + Pandas + Matplotlib
Hardware	Not required
Primary Output	Behaviour fingerprinting, anomaly detection, experiments, graphs, and report

Project concept: Different IoT devices develop characteristic communication patterns. The simulation learns a device's normal profile and identifies significant deviations from that profile.

Prepared as a practical, moderate-scope project plan

1. Project Overview

The proposed project develops a simulation-based system that generates synthetic network traffic for IoT devices, establishes a normal behavioural fingerprint for each device, introduces controlled anomalous behaviour, and identifies deviations using statistical anomaly scoring.

The project is intentionally designed as a moderate two-day study. It does not require physical IoT hardware, live network attacks, deep learning, reinforcement learning, or cloud deployment. The main work is simulation, statistical analysis, visualization, and interpretation.

Core workflow

NORMAL DEVICE BEHAVIOUR → BEHAVIOURAL FINGERPRINT → SIMULATED ANOMALY → ANOMALY SCORE → CLASSIFICATION → EVALUATION

2. Problem Statement

IoT networks contain heterogeneous devices with different communication characteristics. A smart bulb, camera, smart lock, meter, and television can each have a different normal pattern of packet generation and communication destinations.

If a device is compromised, malfunctioning, or otherwise behaving abnormally, its communication pattern may change. The project investigates whether a device-specific statistical fingerprint can detect such changes in a controlled simulation.

Research question

> **Can statistical behavioural fingerprints of simulated IoT devices identify deviations from their normal communication patterns?**

3. Objectives

- Create a synthetic IoT environment containing multiple device types.
- Generate normal communication traffic with device-specific characteristics.
- Construct a behavioural fingerprint for each simulated device.
- Introduce controlled anomalous communication patterns.
- Calculate statistical deviations from the normal profile.
- Classify observations as normal or anomalous using a defined threshold.
- Evaluate detection performance using quantitative metrics.
- Visualize normal behaviour, anomalies, anomaly scores, and detection results.

4. Simulation Architecture

Stage	Function
1. Simulated IoT devices	Create virtual devices representing different IoT categories.
2. Traffic generation	Generate packet rate, packet size, destination count, protocol, and timing data.

Stage	Function
3. Normal profile	Estimate normal mean and standard deviation for each device and feature.
4. Anomaly injection	Introduce controlled changes such as traffic spikes or destination expansion.
5. Detection	Calculate statistical deviation and an overall anomaly score.
6. Evaluation	Compare detected anomalies with the known injected anomalies.
7. Visualization	Produce graphs and tables for interpretation.

5. Simulated IoT Environment

A compact environment of 25 virtual devices is recommended: five device categories with approximately five devices in each category.

Device Type	Normal Packet Rate	Destination Count	Protocol
Smart Bulb	5–10 packets/min	1–3	MQTT
Smart Camera	100–130 packets/min	2–4	HTTP
Smart Lock	8–12 packets/min	1–3	MQTT
Smart Meter	10–15 packets/min	1–2	MQTT/HTTP
Smart TV	20–40 packets/min	2–5	HTTP

Important: These are synthetic simulation ranges and should not be presented as measured characteristics of real commercial devices.

6. Synthetic Data Model

- **Device ID** – unique identifier for the simulated device.
- **Timestamp** – simulated observation time.
- **Packet rate** – number of packets generated per unit time.
- **Packet size** – synthetic packet size.
- **Destination count** – number of communication destinations.
- **Protocol** – simulated protocol label.
- **Connection duration** – duration of the simulated communication session.

Example record

D07 | 10:03 | 7 packets/min | 325 bytes | 2 destinations | MQTT | 4.1 s

7. Behavioural Fingerprinting Method

For each device, normal traffic is used to estimate the mean and standard deviation of selected features. These statistics form a device-specific behavioural fingerprint.

For a measured feature x, the standardized deviation is:

$$Z = (x - \mu) / \sigma$$

An overall anomaly score can combine the absolute standardized deviations of selected features:

$$A = (|Z_{packet}| + |Z_{destination}| + |Z_{size}|) / 3$$

A threshold such as $A \geq 3$ may be used for classification. Thresholds of 2, 3, and 4 can be compared as a small sensitivity experiment.

8. Anomaly Scenarios

Scenario	Normal Behaviour	Injected Anomaly
Traffic spike	Device remains within its usual packet-rate range.	Packet rate is increased substantially.
Destination expansion	Device communicates with a small number of destinations.	Destination count is increased to an unusually large value.
Combined anomaly	Normal packet rate and destination behaviour.	Packet rate and destination count are simultaneously increased.

9. Experimental Design

Experiment	Configuration	Purpose
A — Normal	25 devices; no injected anomaly	Establish baseline behaviour and false-positive behaviour.
B — Traffic spike	Selected devices receive increased packet rates	Test detection of rate deviations.
C — Destination anomaly	Selected devices contact many more destinations	Test destination-based detection.
D — Combined	Packet rate and destination count are changed	Test multi-feature anomaly detection.
E — Threshold study	Threshold = 2, 3, 4	Observe the effect of threshold selection.

10. Expected Visualizations and Results

- **Packet rate vs time** – normal behaviour followed by an injected traffic spike.
- **Destination count vs time** – stable normal behaviour followed by anomalous communication.
- **Anomaly score vs time** – anomaly score crossing the selected threshold.
- **Device-wise anomaly count** – comparison of flagged observations.
- **Detection-performance chart** – TP, FP, and related metrics.
- **Threshold comparison** – effect of thresholds 2, 3, and 4.

11. Evaluation Metrics

Because the injected anomalies are known in the simulation, detected results can be compared with the simulation ground truth.

TP — anomalous observation correctly detected.
TN — normal observation correctly classified.
FP — normal observation incorrectly flagged.
FN — injected anomaly not detected.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Results should be presented as simulation results rather than as evidence of real-world IoT security performance.

12. Three-Person Team Division

The team is divided into three distinct technical roles. Each member owns one major part of the project and contributes on both days, preventing the work from becoming concentrated on a single person.

MEMBER 1	SIMULATION LEAD
Primary responsibility	IoT device modelling, traffic generation, and anomaly injection.
DAY 1 — WORKING TRAFFIC GENERATOR	Build the simulated IoT devices, generate normal traffic, inject the three anomaly types, and export the normal/anomalous CSV datasets.
DAY 2 — VALIDATED SIMULATIONS	Validate the datasets, run the final simulation cases, generate simulation figures, and support final system integration.

MEMBER 2	DETECTION LEAD
Primary responsibility	Behavioural fingerprint calculation, anomaly scoring, and classification.
DAY 1 — WORKING DETECTOR	Implement device-specific fingerprints, calculate statistical deviations, build the anomaly score, and create the initial normal/anomaly classifier.
DAY 2 — FINAL DETECTION RESULTS	Run threshold experiments and calculate TP, TN, FP, FN, accuracy, and detection rate.

MEMBER 3	ANALYSIS & VISUALIZATION LEAD
Primary responsibility	Experimental design, visualization, interpretation, and documentation.
DAY 1 — EXPERIMENT FRAMEWORK	Define the experiments, prepare the analysis workflow, and create the initial plots for normal and anomalous behaviour.
DAY 2 — FINAL ANALYSIS & REPORT	Generate final plots and tables, interpret the results, write the discussion, and integrate the final report and presentation.

Team workflow

MEMBER 1 → BUILDS THE SIMULATION | MEMBER 2 → DETECTS THE ANOMALIES | MEMBER 3 → ANALYSES & PRESENTS THE RESULTS

13. Detailed Two-Day Work Plan

DAY 1 — Development and Integration

Time	Member 1 — Simulation	Member 2 — Detection	Member 3 — Analysis
09:00–10:00	Define device parameters.	Define detection features and scoring approach.	Define experiments and required outputs.
10:00–12:00	Build traffic generator.	Implement fingerprint calculation.	Prepare data-analysis and plotting framework.
12:00–13:00	Test generator.	Test fingerprinting.	Initial validation and data checks.
14:00–16:00	Generate normal traffic.	Implement anomaly score.	Create initial plots.
16:00–18:00	Inject anomalies.	Test detector.	Compare normal and anomalous cases.
18:00–19:00	Export datasets.	Verify detection results.	Review preliminary results.
19:00–20:00	Team integration.	Team integration.	Team integration.

DAY 1 TARGET
By the end of Day 1, the team should have a working end-to-end prototype: traffic generator → behavioural fingerprint → anomaly detector → preliminary plots.

DAY 2 — Experiments, Results and Documentation

Time	Member 1 — Simulation	Member 2 — Detection	Member 3 — Analysis
09:00–10:00	Validate datasets.	Validate detector.	Prepare final figure templates.
10:00–12:00	Run final simulations.	Run threshold experiments.	Generate final plots.
12:00–13:00	Check injected anomalies.	Calculate performance metrics.	Build results tables.
14:00–16:00	Support integration.	Finalize algorithm.	Write results and discussion.
16:00–17:00	Methodology support.	Technical explanation.	Report assembly.
17:00–18:00	Prepare PPT figures.	Prepare PPT methodology.	Prepare PPT results.
18:00–19:00	Final testing.	Final testing.	Final formatting.
19:00–20:00	Presentation practice.	Presentation practice.	Presentation practice.

DAY 2 TARGET
Final simulations, validated detection results, graphs, tables, discussion, report, and presentation ready for submission.

14. Software and Technical Requirements

Component	Recommended choice
Programming language	Python 3.x
Numerical computation	NumPy
Data handling	Pandas
Visualization	Matplotlib
Optional ML library	scikit-learn, only if required
Data format	CSV
Hardware	Not required

15. Recommended Final Report Structure

- Introduction
- Problem Statement
- Objectives
- Proposed Methodology
- Simulation Environment
- Behavioural Fingerprinting
- Anomaly Injection Scenarios
- Detection Method
- Experimental Results
- Performance Evaluation
- Discussion
- Limitations
- Conclusion
- References

16. Scope Control and Limitations

To keep the project achievable within two days, the implementation intentionally excludes physical hardware, live network capture, real attacks, deep-learning models, reinforcement learning, cloud deployment, and a complex web interface.

The traffic and anomalies are synthetic. Therefore, the results demonstrate the behaviour of the proposed method under the chosen simulation parameters and should not be presented as evidence of real-world IoT security performance.

17. Expected Final Deliverables

- Python source code for synthetic IoT traffic generation.
- Synthetic normal and anomalous datasets in CSV format.
- Device-specific behavioural fingerprints.
- Statistical anomaly detection module.
- Threshold comparison results.
- At least four meaningful plots and one results table.
- Detection-performance metrics.
- Final project report.
- Short presentation explaining methodology and results.

18. Final Project Summary

This project provides a moderate-scope simulation study combining IoT, cybersecurity, statistics, data generation, visualization, and experimental analysis. The three-person division gives each member a distinct technical responsibility: Member 1 builds the simulation, Member 2 develops the detection system, and Member 3 performs analysis and visualization. The two-day schedule first establishes a working end-to-end prototype and then focuses on experiments, evaluation, interpretation, and final documentation.

> **Recommended final scope:** 25 simulated IoT devices, 5 device categories, 100–200 simulated observations per device, 3 anomaly types, statistical fingerprinting, three detection thresholds, quantitative evaluation, and 4–6 visualizations.